

EXPLORATION · 27AUG2026

# High level Project Overview

Exploration of AI-assisted reverse-engineering of  
Pharmaverse Admiral functions for development  
of Safety-related ADaM datasets.

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Focus: **1:1 Admiral-function SAS-macro-analogues** vs published pharmaverseadam (PVA).

File: SAS/docs/part\_b\_what\_was\_done\_slides.html

# The Gap

July 31 framing — why SAS analogues still matter after the open-source shift

## Instability is the main issue

- Large pharma moved SAS → open source. Open source has issues; the **main issue is instability**.
- Real possibility **R is superseded by Python** for a future Pharmaverse.
- An R program written today may not run months later.
- **Stability is required** — marketing-application code may need to be re-run *years after filing* for late questions.

## Cost — and who stays on SAS

- R is nominally free; **associated R costs are non-trivial** (GxP qualification, upkeep).
- Smaller pharma / biotech may prefer **SAS in-house**.
- And for **QA of CRO-generated** programming — the Sponsor still owns the submission.



# This exploration

## What was built

- Reusable SAS Macro analogues of R Admiral functions
- Safety ADaM: ADSL ADAE ADVS ADEG ADCM ADLB
- QC against published PVA — a gold standard industry usually does not have

## How we got here

- Early phase (once): more global, Tidyverse-heavy, **not** 1:1. Lesson: 1:1 function analogue is better.
- Without a 1:1 analogue of the *same admiral call*, independent SAS can quietly implement a **different rule** (CTCv5 vs v4, different EG source, different TE flag).
- The risk is **silent semantic drift**. Proof is QC vs PVA *and* the source-data / standards caveats we found.

Learned: standards drift over time changes ADaM content (BRTHDTC, OCCDS vs older ADCM, CTCAE v4 vs v5). ODA is student / academic, not production — some SAS work-arounds exist because of it (e.g. no DATALINES in macros). This is not “an ODA project.”

# Architecture

PVA gold standard: pharmaversesdtm → admiral templates → pharmaverseadam

SAS ADaM: SDTM (read-only) → %m\*\_pva / %m\*\_admiral\_mirror  
→ SAS\_mirrored\_Admiral\_safety\_ADaM/SAS/adam/

QC:

Layer 1 keyed merge

Layer 2 PROC COMPARE digest (PASS / POLICY / FAIL)

Layer 3 traceability (as-programmed vs template steps)

PVA and SAS do **not** share one EG file. PVA used pharmaversesdtm::eg; SAS uses Track A raw.eg.



# Three QC layers

31 July Layer 3 was improved traceability — restored here after Create ADaM

## Layer 1 — keys

Merge SAS vs PVA on analysis keys.  
Orphans, row counts, chkvars.

## Layer 2 — PROC COMPARE

Value digest: PASS / POLICY / FAIL.  
Date-scale \*DTM normalized on working  
copies only.

## Layer 3 — traceability

As-programmed notes vs admiral  
template steps.  
Bridge to a later SAP / spex sheet.

admiral has function docs and templates — **not** a CDISC-style ADaM spec spreadsheet. PVA is generated from those templates.



# QC digest — latest run

| Dataset | n (both) | Orphans | PROC COMPARE |
|---------|----------|---------|--------------|
| ADSL    | 306      | 0       | PASS         |
| ADAE    | 1191     | 0       | PASS         |
| ADVS    | 65032    | 0       | PASS         |
| ADLB    | 83652    | 0       | PASS         |
| ADCM    | 7510     | 0       | POLICY       |
| ADEG    | 78756    | 0       | FAIL         |

ADCM = AENDT (1 understood impute). ADEG = two EG files — SDTM copies stay verbatim. Left as FAIL on purpose.

# Dates — SAS vs R

## QC pairing — resolved

- %m\_qc\_norm\_dtm\_scale on PVA working copies
- Date-scale \*DTM → midnight datetime
- ADAE ASTDTM / AENDTM / LDOSEDTM were a PVA export / Date-collapse artifact
- ADSL + ADAE digest **PASS**

## PVA storage — policy

- PVA still stores date-scale DTM in the XPT
- Not a remaining SAS builder bug
- Do not rewrite SAS datetime to chase the file

Epoch (1960 vs 1970) was an early trap. Compare calendars, not raw numbers, once scale is aligned.



# Closed domains

## ADSL · ADAE · ADVS · ADLB

Full PROC COMPARE PASS.

Counts, keys, analysis values match PVA.

## ADCM

One AENDT row: PVA year-end without  
EOS cap vs admiral

max\_dates=DTHDT, EOSDT.

Keep SAS. Digest already POLICY.

## ADEG

Analysis vars pair. Leftover EGSTRESU /  
EGTPT from two EG files.

FAIL is the story — do not recode.



# Odd rules (template-correct)

| Topic              | What the template does                                                                                          |
|--------------------|-----------------------------------------------------------------------------------------------------------------|
| SCREEN AVISITN     | Blanks SCREEN / UNSCHED / RETRIEVAL / AMBUL. Key repeats — tiebreak *SEQ. Filling VISITNUM would orphan vs PVA. |
| PARAM units        | Unit lives in PARAM (e.g. Hemoglobin (mmol/L)). ADLB never creates AVALU.                                       |
| QTc · R = Result   | QTCBR / QTCFR already on EG. Bazett/Fridericia use RR in seconds. Not why ADEG FAILs.                           |
| TRTEMFL vs AETRTEM | ADaM IG = TRTEMFL (derive_var_trtemf1). AETRTEM is SUPPAE. PVA ADAE has no AETRTEM.                             |

# Standards drift — not “PVA bugs”

| Topic                | What moved / what we keep                                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------|
| BRTHDTC /<br>BIRTHDT | Dropped from many DM extracts for privacy. PVA still has it. SAS omits the column when DM has none.                         |
| CTCAE v4 vs v5       | PVA / template is v4. Hypokalemia G1=G2 → worse-case G2. Hyponatremia: no G2. HGB UNIT_CHECK uses text in parentheses.      |
| OCCDS on CM          | Newer OCCDS puts TRTEMFL (and AVAL/ADT) on ADCM. PVA / ad_adcm.R uses ONTRTFL / PREFL / FUPFL and keeps CMDOSE. Not ported. |



# The gold standard is not issue-free

## ADEG — source, not code

- PVA: pharmaversesdtm::eg
- SAS: Track A raw.eg
- EGSTRESU beats/min vs BEATS/MIN
- EGTPT label vs code

50K+ paired SDTM copies differ. Recoding would invent SDTM.

## Gold standard vs industry

- PVA as gold standard made analogue development easier than industry, where there is **no gold standard**.
- Industry default was **double independent programming** — discuss diffs.
- A disagreement is not automatically a SAS defect (source mismatch, standards drift).



# As-programmed spex — first cut

- **No pre-specified Admiral programming specification was found** for the PVA gold standard or for the reverse-engineered SAS. Admiral ships function documentation + templates. PVA is generated from those templates. Specs must have existed internally for PVA/template authors — they were not available here.
- First cut: SAS/docs/pva\_as\_programmed\_spex.csv, converted to Excel  
SAS/docs/pva\_as\_programmed\_spex.xlsx — what SAS actually calls (derive\_param\_qtc, derive\_var\_atoxgr CTCv4, derive\_var\_trtemf1, remaining ADSL merge, ...)
- prespec\_analogue = Admiral template step name (not a pre-specified programming spec)
- Layer 3 prints this after Create ADaM and joins it to as-built CONTENTS (BOTH / CODE\_ONLY / REGISTRY\_ONLY)
- Compare of as-programmed / as-executed spex to a pre-specified Admiral programming spec was **planned** but **could not be executed** (no spec found) — that work was **curtailed**. In industry, pre-specified spex would be contrasted with as-programmed / as-executed spex.

# Future — from this exploration

## Golden subjects at scale

AI-assisted massive golden-subject testing. This exploration used golden subjects at much smaller scale than production.

## Trusted prior first

Industry first step: compare new code vs a trusted prior set of macros or R functions — there is usually no published gold standard like PVA.

## SAS + R double programming

Large pharma already on R might adopt SAS + R double programming as done here — plus 1:1 analogues to assist QC.

Next on the spec side: AI reading SAP + HA standards into a spex sheet, then checking as-programmed Layer 3 against it.



# Golden subjects vs Murphy<sup>\*</sup>

- CDISC-pilot / PVA is a small case set. A thin golden-subject panel can miss the rest.
- Remaining surprises here were Murphy-class: ADEG two EG files, HGB unit, CTCv4 vs v5.
- AI uniquely enables **hundreds** of frozen rule cases — the counter-weight. If something can go wrong, it will, at the worst time, in the worst way.

<sup>\*</sup> Murphy's Law: If anything can go wrong, it will go wrong, at the worst possible time and in the worst possible manner.



# Building a comprehensive golden-subject set

Cover the *rule space* — not every possible patient. Avoid infinite combinations; still raise programming QC quality.

## Standard CS methods

Partition factors, then **boundaries**, then pairwise / selected 3-way **covering arrays** (hundreds of cases, not millions). Metamorphic checks when there is no oracle; mutation to prove the suite can fail.

## AI vs the oracle

AI drafts the factor table and fills the array from Admiral docs and past Murphy cases. Freeze SDTM in, expected ADaM, one-sentence rule. Do not invent patients with no expected result.

# Pre-planned vs as-programmed spex

Compare only works if both sheets share the **same columns and the same degree of abstraction**.

## Shared schema

Large-pharma derived-data Excel: derivation, inputs, by-vars, method, cutoff/unit, output var. As-programmed / Layer 3 must emit that grain so SAP + HA + programming spex can be row-matched to as-executed.

## Trusted macros as first oracle

Study-proven SAS macros (or Admiral templates) stand in when there is no published gold standard. First step: new code vs that library, then vs golden subjects. Spec-vs-spec here was planned but curtailed — the schema is what would make it executable in industry.



## TAKEAWAY

# 1:1 Admiral-function SAS-macro-analogues vs published PVA

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ADSL ADAE ADVS ADLB **PASS** · ADCM AENDT **POLICY**

- ADEG **FAIL** is two EG files. Dates resolved for QC pairing. PVA date-scale DTM is policy. Layer 3 traceability is back.

Open: SAS\_mirrored\_Admiral\_safety\_ADaM/SAS/docs/part\_b\_what\_was\_done\_slides.html

Spex: SAS/docs/pva\_as\_programmed\_spex.csv

Teal safety review was prior learning / out of scope — not a deliverable of this exploration.